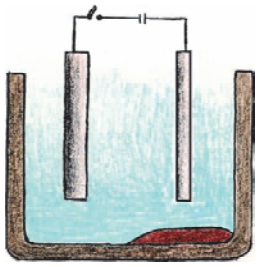
CHAPTER 9

METALS AND METALLURGY



Substances are classified as pure substances and mixtures. We also know that pure substances can either be elements or compounds. Elements (which number more than 100) combine with each other to form all the materials found around us. Elements can be further classified into metals and non-metals. But, is it easy to distinguish between metals and non-metals? In this chapter, we will study some characteristics of elements that help us identify them as metals.

9.1 Some physical properties of metals?

We have already read about the physical properties of metals in previous classes. We know that metals have luster. They can be beaten into thin sheets or foils and even be drawn into thin wires. Metals are conductors of heat and electricity. When we strike a piece of metals, a sonorous sound is produced and usually metals melt at very high temperatures.

Let us do an activity to understand the physical properties of metals in some detail.

Activity-1

From your surroundings, collect samples of some metals, for example - iron, aluminium, copper, zinc. Observe these metals and answer:

- Do all metals have luster?
- Can all metals be beaten into thin sheets or foils?

*(**Note-** Be careful while choosing your samples that you select metals and not alloys).*

We find that these two properties are similar but not identical in metals. Ponder on the following questions about some other properties of metals:

- The wires in different electrical appliances are made of which metal or metals?
- Which metal is used in household wiring?
- Which metal is used to make high tension wires?
- Which metals are used to make cooking utensils and why?

We find that a specific metal is used for a specific purpose. Why don't we use the same metal to solve all our purposes? You must be thinking that there may be more than one reason for selecting a particular metal for a specific task. For example, silver is the best conductor of electricity but due to economic (cost) factors it is not used for wiring in instruments or buildings.

We know that metals are malleable, ductile, conductors of heat and electricity, have high melting and boiling points and produce a sonorous sound. But some of these characteristics are also found in some non-metals, for example- diamond, an allotrope of carbon, has shine and is also a thermal conductor. Another allotrope of carbon, graphite, is grey in colour and a conductor of electricity. Iodine is a non-metal but has shiny crystals. Therefore, it is not right to classify an element as metal or non-metal based on any one physical property.

9.2 Chemical properties of metals

We have seen that there are many similarities, but also some differences, in the physical properties of metals. Do you think that we will see something similar in the chemical properties of metals? We learn a little more about the reactivity of metals by looking at their historical development. Through this study, we try to understand for how long human beings have been using metals and their ease of availability.

Because of differences in the chemical properties of metals, the era when humans were able to obtain them in pure forms is also different (table-1). Let us try to understand what we mean by this statement.

In previous classes, we studied displacement reactions. Can you tell why copper is displaced when we add zinc pieces to copper sulphate solution but zinc is not displaced when we add copper turnings to zinc sulphate solution? We will need to look at some more displacement reactions before we can answer this question.

Table-1: Historical background of obtaining metals

Element	Obtained in	Availability in the earth's crust
Gold	6000 BC	0-00000031%
Copper	4200 BC	0-0068%
Silver	4000 BC	0-0000079%
Lead	3500 BC	0-00099%
Tin	3000 BC	0-00022%
Zinc	2000 BC	0-0078%
Iron	1500 BC	6-3%
Mercury	750 BC	0-0000067%
Platinum	1735	0-0000037%
Cobalt	1739	0-003%
Nickel	1751	0-0089%
Tungsten	1783	0-00011%
Potassium	1807	1-5%
Sodium	1807	2-3%
Calcium	1808	5%
Magnesium	1808	2-9%
Aluminium	1825	8-1%

Activity-2

*(**Note for teachers:-** In a beaker, make a solutions of 2g of copper sulphate in 100 mL water. Similarly, in different beakers make solutions of zinc sulphate, iron sulphate and sodium chloride. The following activity should be carried out in small groups.)*

- Each group of students should take six test tubes and label them as ´A´,´B´,´C´,´D´,´E´, and ´F´.
- In test tubes ´A´ and ´B´ each, take 5 mL copper sulphate solution, in test tubes ´C´ and ´D´ take 5 mL of zinc sulphate solution and in test tubes ´E´ and ´F´ take 5 mL of iron sulphate solution and sodium chloride solution respectively.
- In test tube ´A´ add 2-3 iron nails or awl-pins, in test tube ´B´ add zinc granules, in test tube ´C´ add copper turnings, in test tube ´D´ add a piece of magnesium ribbon and in test tubes ´E´ and´F´ add zinc granules (figure-1).

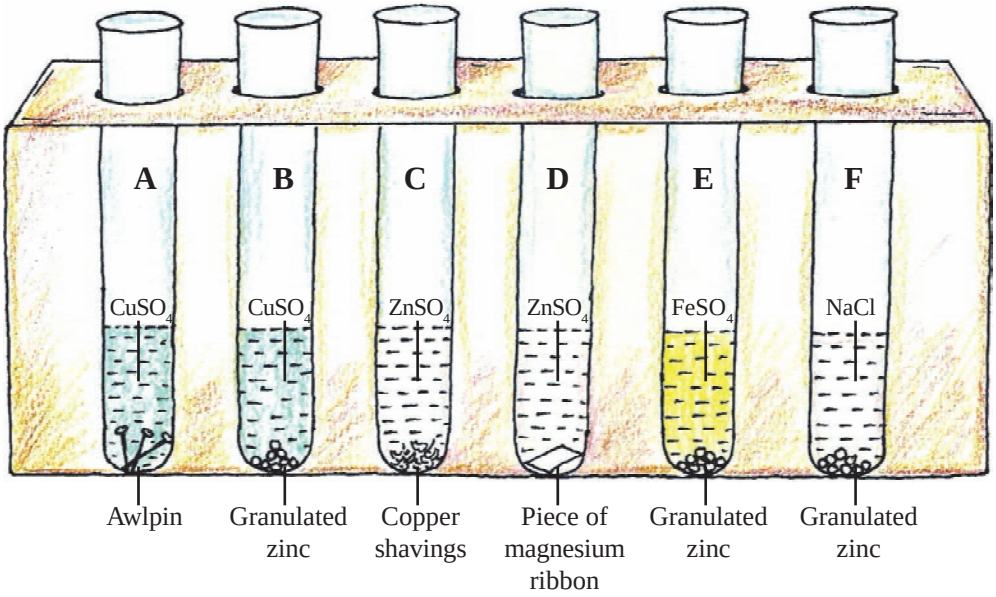


Figure-1: Comparision of reactivity of metals using displacement reaction

- Observe all the test tubes after 5-10 minutes.
  - o Did you notice any change in the appearance of the metals or the solutions.
  - o Is any precipitate being formed?
  - o Can you tell which metal is displacing another metal from its salt solution?

We find that in test tube ´A´, iron is displacing copper from copper sulphate solution, in test tube ´B´, zinc is displacing copper from copper sulphate solution, in test tube ´D´, magnesium is displacing zinc from zinc sulphate solution and in test tube ´E´, zinc is displacing iron from iron sulphate solution.Whereas in test tube ´C´, copper is not displacing zinc from zinc sulphate solution and in test tube ´F´, zinc is not displacing sodium from sodium chloride solution.

Therefore, that metal that displaces another metal from its salt solution is called more reactive of the two. The series obtained when we place metals in descending order of reactivity is known as the activity series. The reactivity of some metals is shown in table-2.

Table-2 : Activity series

Li	Lithium (most reactive)
K	Potassium
Na	Sodium
Ca	Calcium
Mg	Magnesium
Al	Aluminium
Zn	Zinc
Cr	Chromium
Fe	Iron
Sn	Tin
Pb	Lead
H	Hydrogen
Cu	Copper
Hg	Mercury
Ag	Silver
Au	Gold
Pt	Platinum (least reactive)

Decreasing reactivity

Table-2 tells us that lithium is the most reactive metal and platinum is the least reactive metal. It also helps us in comparing two metals, for example, it shows that iron being higher can displace silver from a silver salt solution but silver cannot displace iron from solution of an iron salt. The metals that occur above hydrogen in the activity series can displace hydrogen gas from dilute acids.

If we compare the metals seen in the reactivity series (table-2) with when they were obtained in the pure form (table-1), we realize that the most reactive metals such as sodium, potassium and calcium were obtained in pure form only about 200 years ago whereas the less reactive metals such as gold were known even in the times of ancient civilizations. The reason these metals were used is that they are found in the elemental (pure) form in nature. Very high temperatures are required to obtain pure iron from its ores (hematite and magnetite) and this could happen only after coal and coke replaced wood as fuels. This is why, even though iron is abundantly available in the earth's crust, it was obtained in the pure form only about thirty five hundred years ago.

Mercury, zinc and tin are metals that can be obtained simply by heating their ores. Metals such sodium and potassium, which are above iron, occur as compounds that cannot be reduced to metals by chemical methods.

Let us see how the activity series helps us in understanding the chemical reactions of metals.

9.2.1 What happens when metals are burnt in air?

The more reactive metals - sodium, potassium and lithium - react rapidly with oxygen. When kept in the open, they quickly react with the oxygen present in air to give oxides. Therefore, they are kept under kerosene to prevent them from coming into contact with air. The oxides of sodium and potassium dissolve in water to gives alkalis.

$$4Na + O_2 \longrightarrow 2Na_2O$$

$$Na_2O(s) + H_2O(l) \longrightarrow 2NaOH(aq)$$

Write the balanced equation of the reaction taking place with potassium.

We know that some relatively less reactive metals - for example, magnesium - react slowly with the oxygen present in air but burn rapidly with a white flame when heated. This property of magnesium makes it useful in manufacture of fire-crackers.

Activity-3

Note:- *The bright flame produced during combustion of magnesium ribbon can harm your eyes so be careful.*

- Take a 3-4 inch long piece of magnesium ribbon.
- If there is a white layer on the ribbon then soak it in dilute hydrochloric acid to remove the layer f magnesium oxide.
- Be careful that you do not soak the ribbon in acid for very long or the magnesium will also start reacting with the acid.
- After soaking, carefully blot the ribbon with filter paper and then heat it over a spirit lamp (figure-2).

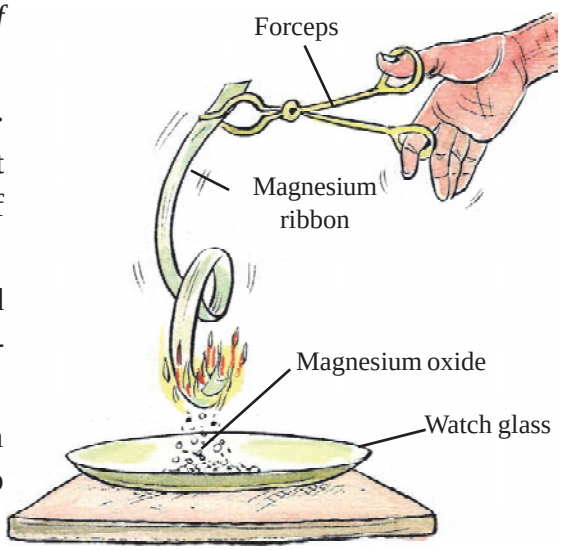


Figure-2: Combustion of magnesium

- Place a moist litmus paper (blue and red separately) near the white smoke seen on burning the ribbon and test its nature.

The white smoke is of magnesium oxide. Is it acidic or basic?

At room temperature, zinc, lead and aluminium metals also react with the oxygen present in air to give their oxides that are deposited as thin layers on the surface of these metals. Once the layer is formed, it prevents further oxidation of the metal.

$$2Zn + O_2 \longrightarrow 2ZnO$$

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

Most metal oxides are basic in nature but some of them such as zinc oxide and aluminium oxide are amphoteric. These oxides react with both acid and bases to give salt and water.

$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH \longrightarrow \underset{\textit{Sodium aluminate}}{2NaAlO_2} + H_2O$$

Copper does not undergo combustion but a black layer of copper oxide gets deposited on the metal. Silver and gold metals do not react with oxygen even at high temperatures.

Questions

1. Why do metals lose their lusterwhen left in the open?
2. Why are gold and silver used in making ornaments and jewellery?
3. Write the balanced chemical equations of the reactions of lead, magnesium, aluminium metals with oxygen?

We saw that potassium, sodium and lithium metals are very reactive. Magnesium reacts more slowly. Let us study some more reactions to understand the reactivity of other metals.

9.2.2 What happens when metals react with water?

Very reactive metals - sodium, potassium and lithium - react with cold water to give metal hydroxide, hydrogen gas and a high amount of heat, therefore, the reactions between these metals and water are exothermic.

$$\underset{\textit{Sodium}}{2Na(s)} + \underset{\textit{water}}{2H_2O(l)} \longrightarrow \underset{\textit{sodium hydroxide}}{2NaOH(aq)} + \underset{\textit{hydrogen}}{H_2\uparrow} + \text{heat}$$

Less reactive metals like magnesium react feebly with cold water but react rapidly with steam or hot water to give magnesium hydroxide and hydrogen gas. The hydrogen gas bubbles formed stich to the surface of the unreacted remaining metal which as a result starts floating on the surface.

$$Mg(s) + 2H_2O(l) \longrightarrow Mg(OH)_2(aq) + H_2\uparrow$$

Metals such as aluminium, zinc and lead react neither with cold water nor with hot water. However, they react with steam to five metal oxide and hydrogen.

$$2Al(s) + 3H_2O(g) \longrightarrow Al_2O_3(s) + 3H_2\uparrow$$

$$2Fe(s) + 3H_2O(g) \longrightarrow Fe_2O_3(s) + 3H_2\uparrow$$

Metals such as gold, silver and copper do not react with water at normal temperature.

9.2.3 What happens when metals react with acids?

We know that metals react with dilute acids to give the corresponding salt and hydrogen gas. Write the balanced chemical equations for reactions of magnesium, zinc and iron with dilute hydrochloric acid.

Do all metals react in this manner? Reactivity towards acids decreases as we go down the activity series from the most reactive metal to the least reactive. Less reactive metals like gold and platinum do not react with dilute acids.

Copper reacts with concentrated nitric acid to give copper nitrate, nitrogen dioxide gas and water.

$$Cu(s) + 4HNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2NO_2(g) + 2H_2O(l)$$

Gold and platinum react only with aqua regia (3 parts concentrated hydrochloric acid and 1 part concentrated nitric acid).

Questions

1. Write the balanced chemical equations for the given reactions:
   (a) Reaction of calcium with water
   (b) Reaction of iron with steam
2. What gas is formed when aluminium metal reacts with dilute hydrochloric acid? Write the equation for the reaction.
3. What is aqua regia?

Now we will see how different metals are obtained, how they can be protected from environmental factors, and how they can be mixed with other metals to increase their usefulness.

9.3 Occurrence of metals

Metals are usually found in the earth's crust. Some metals occur in the free state and others in the combined state. Generally, metals which lie low in the reactivity series - such as gold and platinum - are found in free form whereas, the more reactive metals react with other elements to form compounds and are found in combined form. Solid inorganic substances which occur naturally in the earth's crust, whose chemical composition is fixed and whose physical properties can be estimated, are known as minerals.Metals are extracted from these minerals. Minerals which contain a very highpercentage of a particular metal (and low percentage of impurities) and from which the metal can be easily and profitably extracted, are called ores.Usually, minerals are found in different types of rocks. All ores are minerals but all minerals are not ores.

Table-3: Ores of metals

S.No.	Type of ore	Name of ore	Formula of ore
1	Oxide	Hematite	Fe_2O_3
		Magnetite	Fe_3O_4
		Bauxite	$Al_2O_3.2H_2O$
		Cassiterite	SnO_2
2	Carbonate	Dolomite	$MgCO_3.CaCO_3$
		Limestone	$CaCO_3$
		Calamine	$ZnCO_3$
3	Sulphide	Iron pyrite	FeS_2
		Copper pyrite	$CuFeS_2$
4	Sulphate	Gypsum	$CaSO_4.2H_2O$

9.3.1 Important minerals and their distribution in Chhattisgarh

Chhattisgarh is rich in minerals and holds an important place in our nation's mineral pro duction. In all, 28 different types of minerals are found in Chhattisgarh. It is the only state in India where tin, which is a strategically important mineral, is found. The important minerals found in Chhattisgarh are shown in table-4.

Table-4: Important minerals found in Chhattisgarh

S.No	Name of mineral	Districts where minerals are found
1.	Hematite	Dantewada, Bastar, Kanker, Narayanpur, Raipur, Kabirdham
2.	Bauxite	Sarguja, Korba, Jashpur, Kanker, Bastar, Kabirdham
3.	Limestone	Raipur, raigarh, Janjipur - Champa, Kabirdham, Bastar, Bilaspur, Durg, Rajnandgaon
4.	Dolomite	Durg, Kabirdham, Bilaspur, Janjipur - Champa
5.	Alexandrite (Chyrsoberyl)	Raipur (Devbhog)
6.	Cassiterite	Dantewada, Bastar
7.	Corundum	Dantewada
8.	Gold	Raipur, Kanker, Raigarh, Mahasamund, Bastar
9.	Asbestos	Raipur, Bastar, Sarguja, Jashpur

9.4 Metallurgy

The process of obtaining metals from ores is known as extraction of metals or metallurgy. There are three main steps in this process:

1. Concentration of ore 2. Extraction of metal 3. Purification of metal

Depending on the type of metal and the type of ore, different techniques are used for extraction of metals. Let us study in some detail.

9.4.1 Concentration of ore

Ores obtained from mines are usually contaminated with large amountsof impurities such as soil, sand, etc., called gangue. The impurities mustbe removed from the ore prior to the extraction of the metal; this is known as concentration of ore. Let us learn about different techniques used in concentration of ores.

(1) **Gravity separation method-** When there is a large difference in the densities of the ore and gangue material then gravity separation method is used for concentration of ore. In this, the ore is first finely ground, then shaken under a stream of water which leads to the heavier ore particles settling down and the lighter gangue particles floating to the top. These gangue particles are either separated or get washed away by the water (Figure-3). Generally, this method is used for concentration of carbonate or oxide ores, for example, the ore hematite is concentrated using this technique.

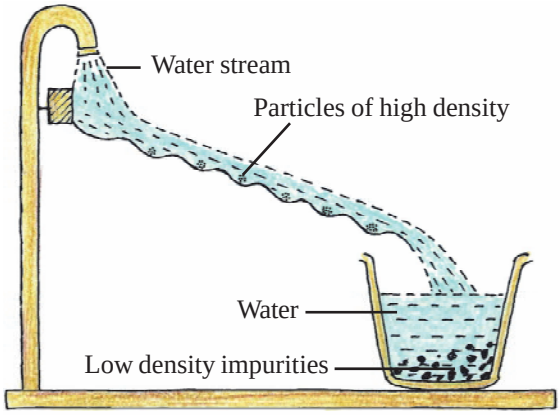


Figure-3: Gravity separation method

(2) **Magnetic separation method-** This method can be used if magnetic substances are present in the ore. In this method, the finely ground ore is passed over a conveyor belt which is stretched over magnetic rollers. The magnetic particles remain stuck to the belt till they are moved away from the magnetic effect of the roller whereas the non-magnetic particles fall from the belt as soon as it starts moving. In this way, two heaps are obtained; one heap is of magnetic material and the other is of non-magnetic material (Figure-4). This technique is used to separate magnetic impurities from tin ore (SnO_2).

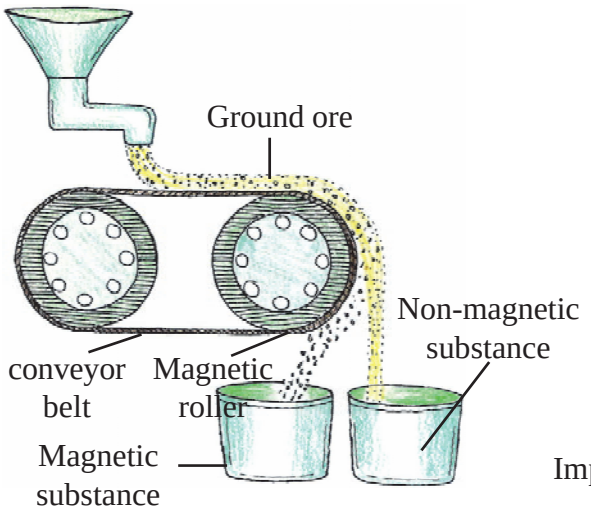


Figure-4: Magnetic separation method

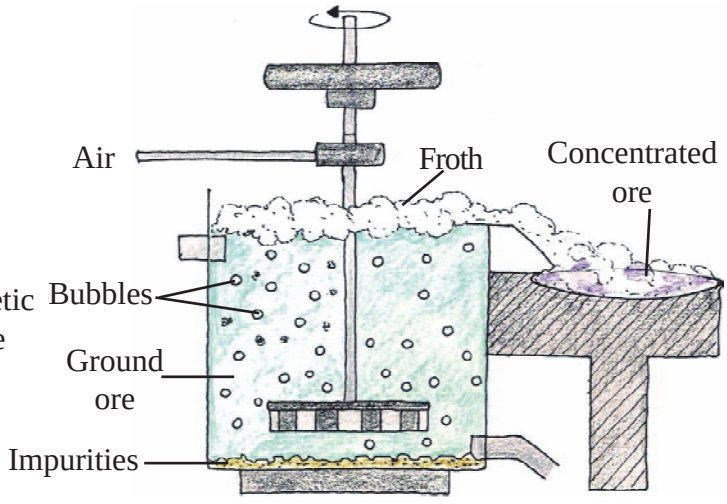
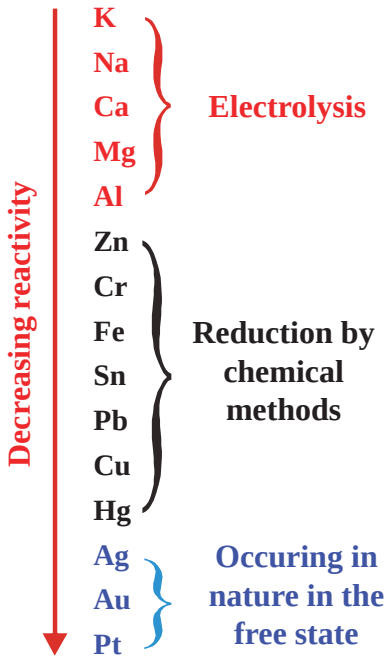


Figure-5: Froth floatation method

(3) Froth flotation process- This method is used for concentration of sulphide ores. In this method, finely ground sulphide ore is mixed with water and pine oil and compressed air is passed through the mixture causing frothing. A layer of pine oil is formed on the surface of the sulphide ore particles which causes the air bubbles to stick to them. The ore particles come to the surface with the froth while impurities like soil, pebbles, stones etc. settle down (figure-5).

Chhattisgarh has high quantities of iron ores; find out the methods used in concentration of these ores.

Table-5: Activity series and related metallurgy



9.4.2 Extraction of metal

The method used for extraction of any metal depends on its reactivity and nature of its ore.

(A) We know that it is difficult to separate the more reactive metals like sodium, potassium and calcium as elements from their compounds. When electric current is passed through these compounds, the cation dissociates from the anion and is reduced to the corresponding metal at the cathode. This process is known as electrolysis. Aluminium is also obtained by electrolysis (table-5).

(B) Medium and less reactive metals are obtained from their oxide, sulphide and carbonate ores by chemical reduction method. Some lesser or low reactive metals like mercury can be obtained simply by heating their oxides or sulphides.

For oxides, sulphides and carbonates of other metals, two steps are followed to easily reduce the metal from its ore. These steps are described below.

9.4.2.1 Converting the metal sulphide or carbonate to corresponding oxide

The method used to convert a metal sulphide or carbonate to metal oxide depends on the impurities present in the ore.

(i) The heating of concentrated sulphide ore a little below its melting point in the presence of air or oxygen is known as roasting. During roasting, the impurities are vaporized which eases the reduction of the metal compound. Roasting is usually carried out for sulphide ores.

$$2ZnS + 3O_2 \xrightarrow{\triangle} 2ZnO + 2SO_2\uparrow$$

(ii) The heating of concentrated ore a little below its melting point in the absence of oxygen is known as calcination. This method is used for carbonate ores.

$$ZnCO_3 \xrightarrow{\triangle} ZnO + CO_2\uparrow$$

On heating, carbonate ores dissociate to release carbon dioxide gas and give metal oxide.

9.4.2.2 Reduction of metal from metal oxides

Once the ores are converted to their oxides, they are then reduced to obtain the respective metals. The metal oxide is heated along with flux and reducing agent and this gives the metal in liquid form. This process is known as smelting. The flux helps in removal of hard to melt impurities. Smelting can be carried out in two ways:

(i) Carbon reduction method - roasted or calcinated ore is heated at high temperature (700°C-1800°C) with coke or carbon in a blast furnace and the oxide is reduced to liquid metal.?

$$ZnO + C \xrightarrow{\triangle} Zn + CO\uparrow$$

(ii) Chemical reduction method - Metals lower in the reactivity series can be displaced from solutions of their ionic salts by metals higher up in the reactivity series. This principle is used to separate metals using the displacement reaction. We have previously seen that copper can be displaced from copper sulphate solution by the addition or iron or zinc powder.

$$CuSO_4 + Fe \longrightarrow FeSO_4 + Cu$$

$$CuSO_4 + Zn \longrightarrow ZnSO_4 + Cu$$

Do you know the various metals which can be extracted using this method?

9.4.3 Purification of metals

The metals obtained by reduction have several impurities and is known as impure metal. The process of obtaining pure metal from impure metal is known as purification of metal. It is carried out by the following two methods:

(i) Physical methods- The two main physical techniques for purification of metals are liquefaction and zone refining (also known as fractional crystallization).

(A) Liquefaction- Metals such as tin, lead etc. that melt easily are placed over a sloping hearth in a furnace. The metal melts and flows off while the impurities are left behind.

(B) Zone refining or fractional crystallization- This method is used for refining of metals such as gold, silver and platinum. Presence of impurities lowers the melting point of a metal. Zone refining is based on the principle that impurities are more soluble in liquid state of a metal as compared to the solid state.

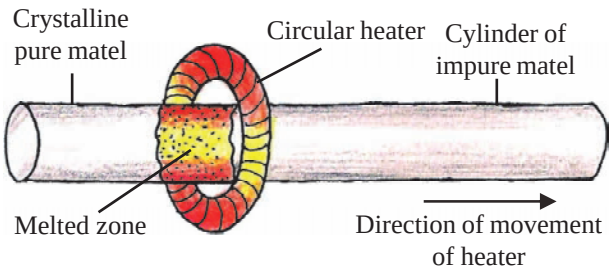


Figure-6: Zone refining

In this method, a cylinder of impure metal is placed within a circular heater till the metal starts melting. A heater is slowly moved from one end of the cylinder to the other.A narrow zone of the metal melts from the heat and the melt containing the impurities moves along with the heater to the other end of the rod while the pure metal solidifies and is left behind. Thus the impurities are concentrated towards one end of the rod and the remaining part is purified. The process is repeated several times to obtain ultra-pure metal. The part containing the impurities is cut off from the rod.

(ii) **Chemical Methods-** some metals are also refined using chemical methods. To obtain ultra-pure metal, electrolysis is used. A thick rod of impure metal is made the positive electrode and a thin rod of pure metal is used as the negative electrode. Solution of salt of the same metal is used as electrolyte.

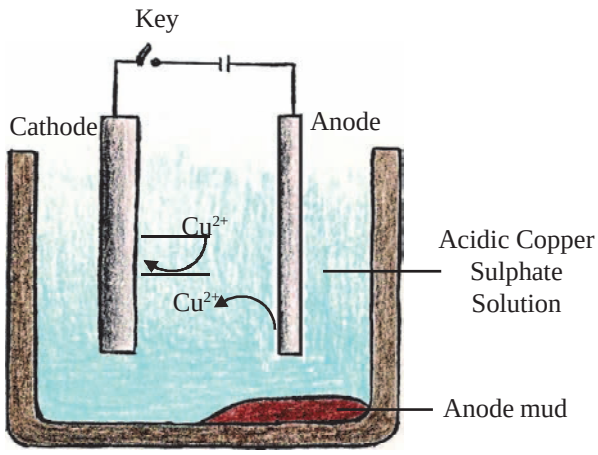
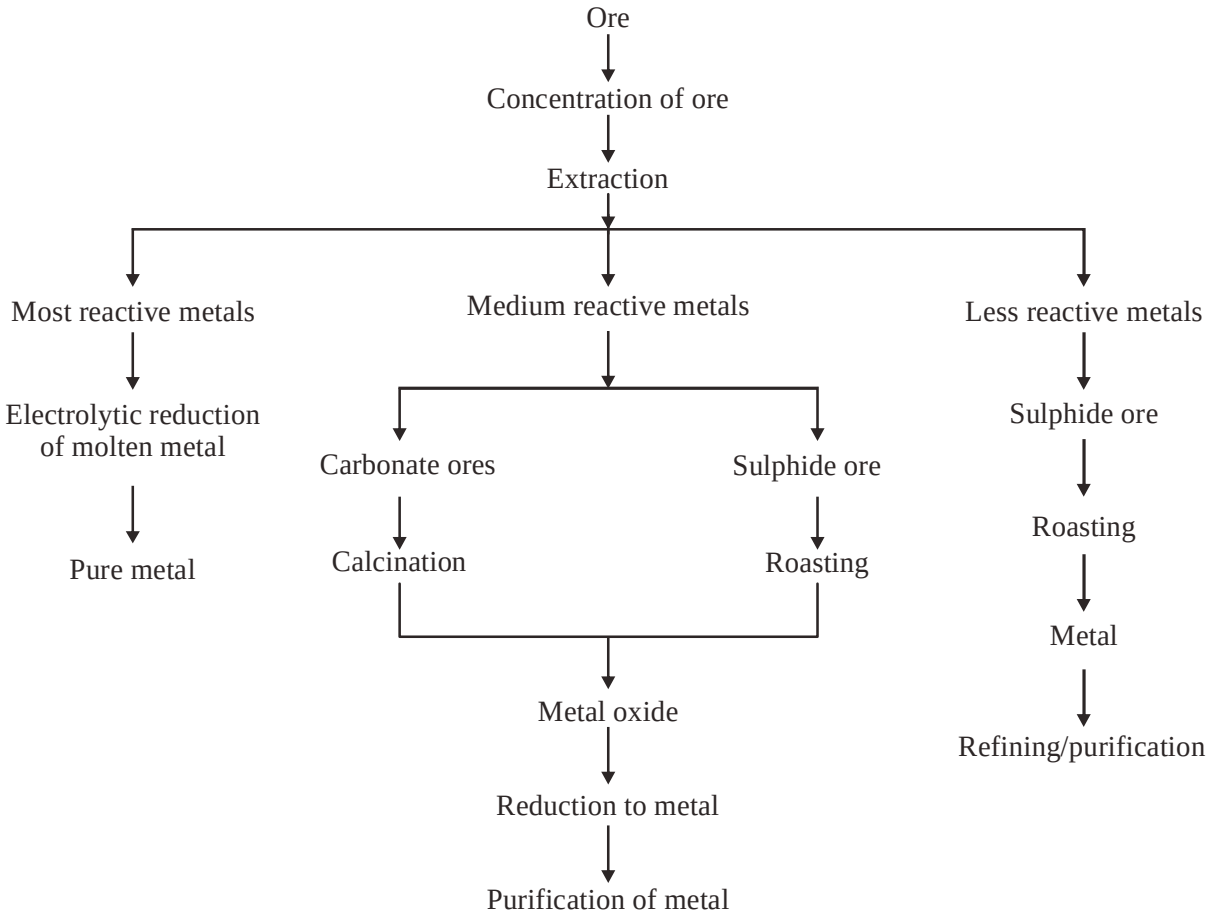


Figure-7: Electrolytic refining of metal

When electric current is passed through the electrolyte then the positive ions present in the solution (metal ions) move towards the negative electrode and are reduced to metal. At the positive electrode, the metal atoms get oxidized and dissolve in the solution. Once they dissolve in the solution these metal ions also move towards the cathode. The impurities in the metal at the anode are usually insoluble. As the metal atoms from the anode enter the solution, the substance at the anode reduces and the impurities collect below the anode. These collected impurities are called anode mud (Figure-7). This technique is used in the purification of chromium, nickel, copper, zinc etc.



Flowchart 1: Metallurgy

Activity-4

Take two copper wires of different thickness, if the wires are coated clean them with sandpaper. Connect the thicker wire to the positive end and thinner wire to the negative end of a cell. Soak both the wires in a solution of copper sulphate ensuring that the wires do not touch each other inside the solution. Observe the solution after some time and answer the following questions:

- Did the copper go from one electrode to the other?
- On which electrode did the amount of copper reduce and on which did it increase?

Similarly, when we use electrolytic refining metal is obtained on the negative electrode.

Questions

1. Write the names and chemical formulae of oxide ores of iron.
2. How are sulphide ores concentrated?
3. Na, K and Ca metals are obtained from their compounds using electrolysis, why?

9.5 Iron

Iron has played a prominent role in the human civilization and the Iron Age was the longest lasting historical era. Iron is an active metal and is found in nature in combination with other elements. Hematite (Fe_2O_3), magnetite (Fe_3O_4), siderite ($FeCO_3$) and iron pyrite (FeS_2) are the important ores of iron.

9.5.1 Metallurgy of iron

Hematite (Fe_2O_3)is the iron ore most commonly found in Chhattisgarh. Once the ore is obtained from the mines, it is crushed into a fine powder using a crusher or stamp mill which is then concentrated using gravitational separation method.

Post concentration, the ore is roasted so that impurities such as arsenic and sulphur present in the ore get converted to their gaseous oxides and are removed. The ore becomes porous facilitating metal reduction.

$$4As + 3O_2 \longrightarrow 2As_2O_3\uparrow$$

$$S + O_2 \longrightarrow SO_2\uparrow$$

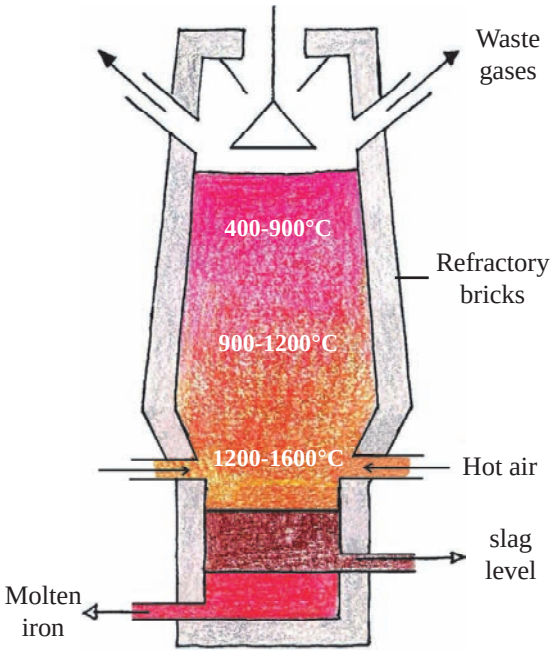


Figure-8: Blast furnace

Mainly sand is found as impurity in the ore which is removed with the help of calcium carbonate during smelting. The oxides of iron are reduced using carbon (coke or anthracite). The process requires high temperatures, about 1800°C. If the process is carried out with frequent breaks then a lot of energy will be wasted in heating the mixture to the required temperature. Therefore, smelting is carried out continuously in a blast furnace (Figure-8).

Blast furnace is a cylindrical furnace made of steel whose insides are lined with refractory bricks. The broad portion in the middle of the furnace has inlets through which hot air (700-800°C) is blown into the furnace. The upper portion of the furnace has an outlet for gases formed in the furnace during the reaction. The lower portion has two outlets - the first higher one for slag and the second one for molten metal.

The roasted ore is mixed with lime stone and coke and fed into the blast furnace from the top and a blast of hot air is blown from the bottom. The temperature throughout the blast furnace does not remain the same so as the ore, lime stone and coke mixture comes down it has to pass through different temperature zones. The following reactions take place in a blast furnace:

Coke reacts with oxygen to form carbon dioxide gas. Since excess of coke is added, therefore the extra coke reacts with carbon dioxide gas to form carbon monoxide. Coke also reacts with insufficient oxygen to form carbon monoxide. This process ensures that high temperature (400°C-900°C) is maintained within the furnace.

$$C + O_2 \longrightarrow CO_2\uparrow$$

$$CO_2 + C \longrightarrow 2CO\uparrow$$

$$2C + O_2 \longrightarrow 2CO\uparrow$$

Oxide ores react with carbon monoxide gas to give iron and carbon dioxide. This reaction is exothermic.

$$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2\uparrow$$

Since this reaction maintains high temperature in the furnace therefore the carbon dioxide gas in not allowed to escape and again sent in to react with coke. A small amount of hematite also reacts with the coke to give iron and carbon monoxide.

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO\uparrow$$

At high temperature in the furnace, lime stone dissociates into calcium oxide and carbon dioxide. Since this is an endothermic reaction therefore limited quantities of lime stone are added to the furnace because excess of limestone can reduce heat and cause energy loss. The calcium oxide formed during the reaction is used to remove impurities from the ore and is known as flux. The flux reacts with silica to form calcium silicate, also known as slag.

$$CaCO_3 \xrightarrow{\triangle} CaO + CO_2$$

$$CaO + SiO_2 \xrightarrow{\triangle} CaSiO_3$$

Flux + impurity *slag*

Since the furnace temperature is kept between 1200°C and 1600°C, iron and calcium silicate remain in molten state. Their densities being different, they collect in two layers at the bottom of the furnace and are removed through outlets at two different layers.

The iron thus obtained has high quantities of coke and is known as pig-iron. Different techniques can be used to refine this iron and this gives iron of different grades such as cast iron and wrought iron.

Metal art in Chhattisgarh

Iron art and craft hold a special place in traditional art-work of Chhattisgarh. In ancient ages, iron was obtained by breaking down iron containing rocks, mixing them with coal and melting the mixture. Even today in some parts of Chhattisgarh, iron is extracted from special types of rocks and used to prepare figurines. But these days, mostly commercial iron is purchased, cut and heated to make decorative items such as knives, tongs, axes, oil-lamp holders, animal-statues, flower-vase etc. Iron art in Bastar focuses on making traditional idols of gods and goddesses for worship. Windows, doors, swings are also made in traditional iron-art.

In Dhokra metal casting art, first a wax figure is created which is then coated with clay and baked in an oven and molten metal is poured inside. The metal takes the place and shape of the wax figure. Decorative figures of bulls, gods-goddesses, horses, elephants etc. are created using this technique. Dhokra art is practiced in Raigarh, Kondagaon, Bastar etc. Door knobs and handles, animal horns, crowns etc. are also made in this art-form.



Figure - Metal art

Questions

1. Which reactions are helpful in maintaining high temperatures within the blast furnace?
2. Which compound is used as flux in the blast furnace during extraction or iron metal?

9.6 Corrosion

We know that active metals - for example, aluminium, zinc - react rapidly with oxygen and moisture and are converted into their respective compounds. This process is known as corrosion. The corrosion of iron by oxygen and water is called rusting. Next, we will discuss the process of corrosion and how to prevent it.

Activity-5

(**Note-** This activity should be carried out in summer or winter but not during the rainy season when due to high humidity it is difficult to get results).

- Take three test tubes and label them 'A', 'B' and 'C' respectively.
- Put 3-4 awl-pins in each test tube. Add enough water in test tube 'A' so that the pins are submerged.
- Moisten a piece of cotton and put it in lowest part of test tube 'B'.
- Leave awl-pins in test tube 'C' untouched.

- Now make the test tubes air-tight by placing balloons on their mouths. Put the test tubes in a test tube stand and place them in a hot place (or sunlight) for 3-4 hours (figure-9). Observe after this time. Remember that rates of reactions are slower in cold weather.
- Did rust form on the awl-pins in all three test tubes?
- Was the amount of rust formed same?
- Did you notice any change in the balloons covering the test tube mouths?

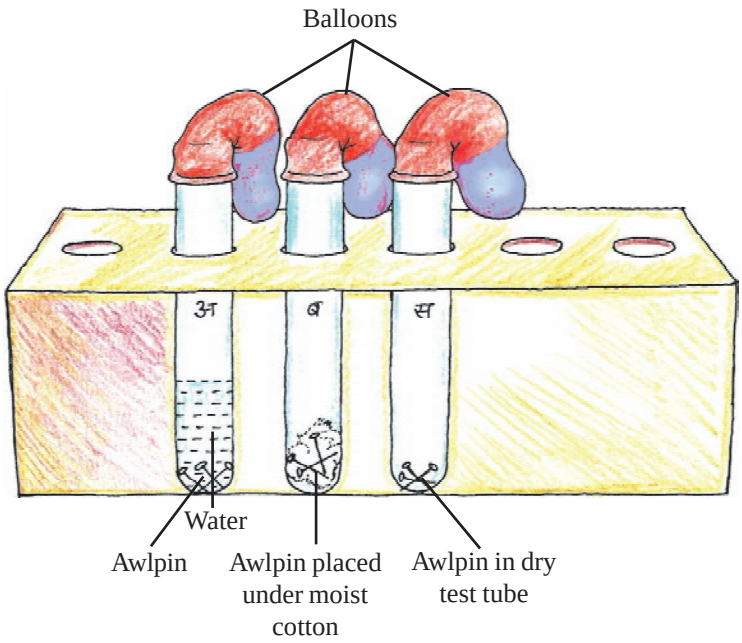


Figure-9: Effect of air and moisture on rusting process

- What could be the cause of this change? Could it be the formation or use of any gas?
- Which test tube among the three ('A', 'B' or 'C') test tubes showed more changes?

Presence of oxygen and moisture is necessary for rusting of iron but other factors such as presence of salts and acids also affect rusting. Their presence increases rate of rusting and this is why iron rusts faster in areas close to seas and oceans.

Nature of metal also affects the rate of corrosion (formation of oxide layer on metal surface). Most corrosion prevention metals try to prevent air and moisture from coming in contact with the metal. The process of applying a protective layer, usually paint or grease, on iron surface to prevent contact with atmosphere is known as barrier protection. Electroplating is used to coat the surface of iron with metals such as tin, nickel and chromium.

Activity-6

- Make a list of iron objects in your home and school and note which of them are coated with paint and which are coated with grease.
- List the basis of selection for any one of the two barriers.

Corrosion of metal can also be prevented by using a more reactive metal. This process is known as sacrificial protection. You might have heard of galvanized iron where the iron object is coated with a thin layer of zinc metal. Zinc being more reactive than iron, reacts more readily with oxygen and a thick layer of ZnO is formed which prevents the rusting of iron.

- Some metals are protected by applying a corrosion-resistant solution. In this process basic solutions of chromate or phosphate are used as corrosion resistant solutions.
- Sometimes, iron is mixed with other metals to form alloys and these do not rust at all or rust rarely.

9.6.2 Chemical principles of rusting of iron

The process of corrosion is actually an electrochemical process. When metals like iron are placed in air and humid conditions then negatively and positively charged regions develop on its surface because of which it starts behaving like an electrochemical cell.

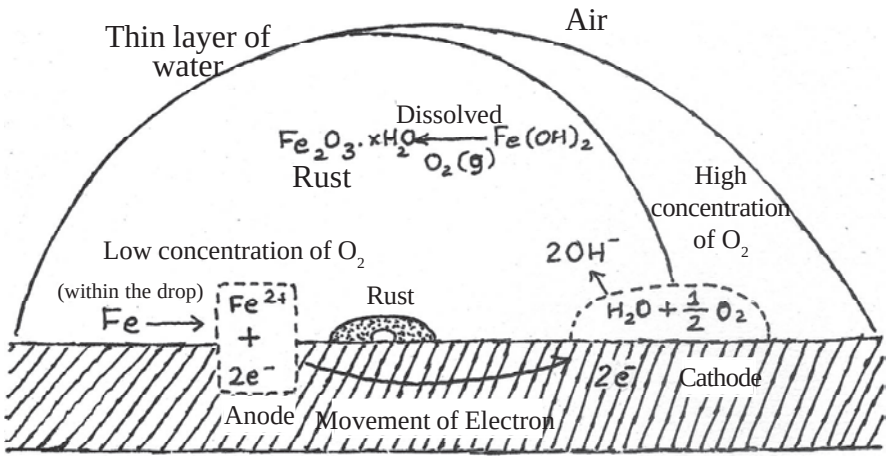


Figure-10: Chemical reactions in rusting of iron

Anode -Fe moves into the solution as Fe^{2+}

$$Fe \longrightarrow Fe^{2+} + 2e^-$$

Cathode - In the presence of oxygen, these electrons are taken up water molecules to form OH- ions.

$$2H_2O + O_2 + 4e^- \longrightarrow 4OH^-$$

The Fe^{2+} ions formed on the anode react with the OH^- to form $Fe(OH)_2(s)$. This iron hydroxide reacts with the oxygen in the atmosphere in the presence of moisture to give hydrated iron oxide.

$$2Fe(OH)_2 + \tfrac{1}{2}\, O_2(g) + H_2O(l) \longrightarrow Fe_2O_3.xH_2O(s)$$

This hydrated iron oxide is known as rust.

9.7 Alloys

Alloying is a popular method of improving the properties of a metal. A homogeneous mixture in which at least one of the primary components is a metal is called an alloy. Alloys are examples of solid solutions. Human beings have been making and using alloys since prehistoric times. Alloying is done to increase hardness, improve quality or decrease melting point. Alloys can be easily cast or moulded into different shapes and find uses in many areas.

An alloy is prepared by first melting one of the metals, and then,dissolving the other metals or non-metals in it in definite proportions. The mixture is then cooledto room temperature to get alloy. While making alloys, metals (or other elements) are chosen carefully so that we can get the desired properties in the resulting alloy.

Iron isthe most widely used metal but we can't use it in its pure form because pure iron is very relatively soft. But, if a small amount of carbon is mixed in it, it becomes hard. Stainless steel is an alloy of iron where nickel and chromium are also added along with carbon. Chromium prevents iron from rusting thus increasing its usefulness.

Table-6 : Some useful alloys

S. No.	Alloy	Constituents and their percentages	Uses
1.	Brass	Cu - 60-80%, Zn - 40-20%	Utensils and household objects
2.	Bronze	Cu - 75- 90%, Sn - 25-10%	Coins, utensils, statues
3.	German Silver	Cu - 50%, Zn- 25%, Ni - 25%	Coins, utensils, and ornaments
4.	Magnalium	Mg - 5% , Al - 95%	Structural framework of aircrafts
5.	Bell metal	Cu - 80% , Sn - 20%	For making bells
6.	Gun metal	Cu - 87% , Zn - 3%, Sn -10%	For making material used in Engineering
7.	Solder	Sn - 50%-75%, Pb - 50-25%	For joining together metal wires
8.	Duralumin	Al-95%, Cu-4%, Mg-0.5% Mn-0.5%	Manufacture of aircrafts, helicopters, jet planes etc
9.	Stainless steel	Cr-11.5%, Ni- 2 %, C- trace amount, remaining -Fe	Manufacturing knives, cooking utensils etc.
10.	Alnico	Al-12 %, Co-5 %, Ni-20%, Fe	Making permanent magnets

Activity-7

- Make a list of all objects made of metal that are used in your schools or at home.
- Note which of the objects is made of pure metals and which is made of alloys. Think: what could be the reason for using pure metal or alloy while making a particular object.

Chhattisgarh is a primary site for iron ore and also has many steel production plants. Chhattisgarh has a long history of working with iron and different areas follow different traditions of iron craftsmanship. Our ancestors not only discovered techniques of iron extraction but also prepared many different alloys by mixing iron with other metals.

Questions

1. How will the rate of corrosion be affected in water that has a high concentration of salts?
2. Why do we make alloys?
3. What is galvanization?

Keywords

Mineral, ores, metallurgy, extraction, roasting, calcination, gravity separation method, froth floatation method, magnetic separation method, liquefaction, smelting, flux, blast furnace, refining, corrosion, barrier protection, sacrificial protection, alloys



What we have learnt

- Metals have metallic luster, ductility, malleability, sonority and they are good conductors of heat and electricity.
- When metals are arranged in descending order of their reactivity, we get the reactivity series.
- More reactive metals can displace less reactive metals from the latter's salt solutions.
- Metals react with oxygen to form metal oxides which can be basic or amphoteric in nature.
- Metals react with water to release hydrogen gas.
- Metals react with acids to give the corresponding salt and hydrogen gas.
- Metals are present in nature in both free and combined forms.
- The process of separating metals from their ores is known as metallurgy. Metallurgy has three main steps - (1) concentration of ore (2) extraction of metal and (3) purification of metal.
- The method of concentration depends on the nature of the metal ore.
- Concentration of ore is done mainly using three methods - gravity separation method, froth floatation method and magnetic separation method.
- The heating of concentrated sulphide ore a little below its melting point in the presence of air or oxygen is known as roasting.The heating of concentrated ore a little below its melting point in the absence of oxygen is known as calcination.
- Reduction of metal from its metal oxide is done mainly using coke or charcoal.
- Flux is the substance added to remove molten impurities from an ore. Flux reacts with the impurities to form slag.
- Reactive metals react with oxygen and moisture and convert into their compounds (usually oxides); this process is known as corrosion.
- Metals are mixed with other metals and non-metals to make alloys.

Exercises

1. Choose the correct option–

   (i) Which among the following metals is most reactive–

   (a) Mg (b) Al

   (c) Na (d) Zn

   (ii) The formula of bauxite ore is–

   (a) $MgCO_3$ (b) $Al_2O_3.2H_2O$

   (c) Fe_2O_3 (d) SnO_2

   (iii) Which method among the following is used in concentration of iron pyrite–

   (a) Magnetic separation (b) Gravity separation

   (c) Froth floatation (d) Electrolysis

   (iv) Calcium silicate is–

   (a) Alloy (b) Gangue (c) Slag (d) Flux

   (v) The constituent metals in stainless steel are–

   (a) Copper, zinc and tin (b) Iron, chromium and nickel

   (c) Copper, iron and zinc (d) Iron, tin and aluminium

2. Fill in the blanks

   (i) The process of removal of a less active metal from its salt solution by a more reactive metal is known as …….. (displacement/precipitation) reaction.

   (ii) The primary iron ore found in Chhattisgarh is …….. (hematite/magnetite).

   (iii) Alloys are a ……(homogeneous/heterogeneous) mixture of two or more metal or metals and non-metals.

   (iv) The process of heating an ore at a temperature a little below its melting point in the absence of oxygen is called …… (roasting/ calcination).

3. Explain the following

   (i) Ore (ii) mineral (iii) slag (iv) flux

4. Write the balance chemical equations for the given reactions:

   (i) Reaction of aluminium metal with steam.

   (ii) Reaction of zinc oxide with sodium hydroxide.

   (iii) Reaction when calcium carbonate is heated.

   (iv) Reaction of oxygen with sodium.

5. Explain the process of reduction of metal from a metal oxide.

6. Discuss the different methods used to prevent corrosion.

7. In the purification of a metal (M), what all will act as anode, cathode and electrolyte. Explain with an example.
8. Explain the following techniques used in concentration of ore:
   (i) Gravity separation method
   (ii) Froth flotation method
   (iii) Magnetic separation method
9. Discuss the extraction of iron from hematite as per the following steps:
   (i) Concentration of ore
   (ii) Reduction of oxide to metal (along with chemical equation)
   (iii) Diagram of blast furnace
10. Explain the electrochemical principle behind corrosion.
11. Give reasons for:
   (i) Gold, silver and platinum are used in making jewelry
   (ii) Sodium, potassium and lithium metals are stored under kerosene
   (iii) Although aluminium is an active metal it is still used to prepare cooking utensils
   (iv) Carbonate and sulphide ores are converted into their oxides for extraction of metal